Kinematics is the branch of mechanics which deals with the motion of objects without taking force into account. The Greek word ***"kinema"*** means ***"motion".***

2.2 CONCEPT OF REST AND MOTION

The concept of rest and motion can be well understood by the following elucidation (Figure 2.1). A person sitting in a moving bus is at rest with respect to a fellow passenger but is in motion with respect to a person outside the bus. The concepts of rest and motion have meaning only with respect to some reference frame. To understand rest or motion we need a convenient fixed reference frame.

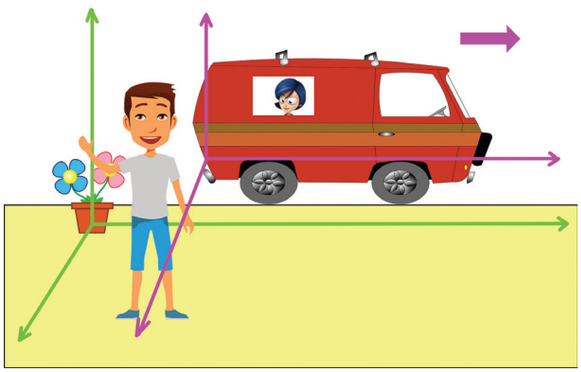


Figure 2.1 Frame of Reference

Frame of Reference:

If we imagine a coordinate system and the position of an object is described relative to it, then such a coordinate system is called frame of reference.

At any given instant of time, the frame of reference with respect to which the position of the object is described in terms of position coordinates (x, y, z) (i.e., distances of the given position of an object along the x, y, and z–axes.) is called **"Cartesian coordinate system"** as shown in Figure 2.2

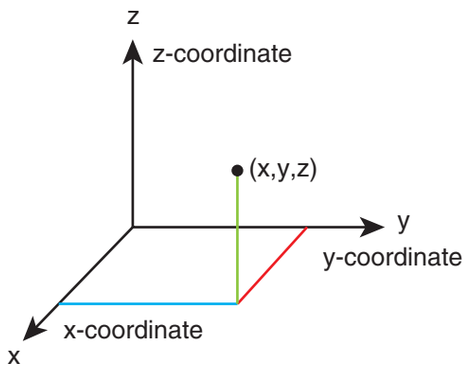


Figure 2.2 Cartesian coordinate system

It is to be noted that if the *x, y* and *z* axes are drawn in anticlockwise direction then the coordinate system is called as "right–handed Cartesian coordinate system". Though other coordinate systems do exist, in physics we conventionally follow the right–handed coordinate system as shown in Figure 2.3.

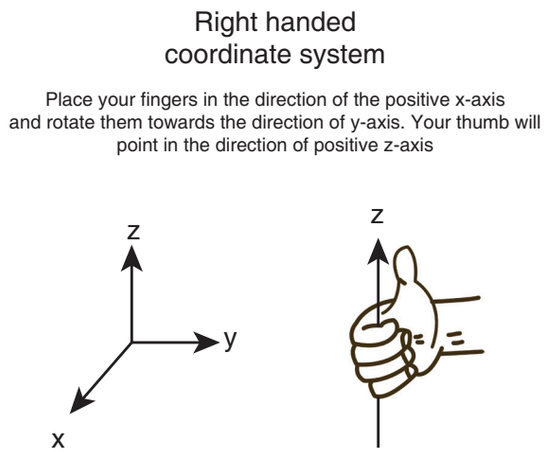


Figure 2.3 Right handed coordinate system

The following Figure 2.4 illustrates the difference between left and right handed coordinate systems.

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